KAIZEN MATHS

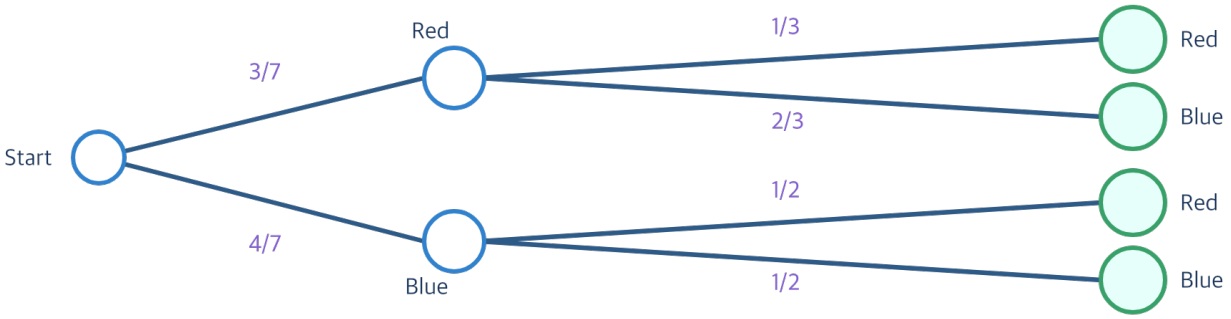
Lesson starter

Tree Diagrams and Conditional Probability

Conditional Trees

Name _____

SA1



A bag contains 3 red counters and 4 blue counters. Two counters are drawn without replacement. Find P(two blue counters).

STA-3EBB-L2 · SA

Show your working

KAIZEN MATHS

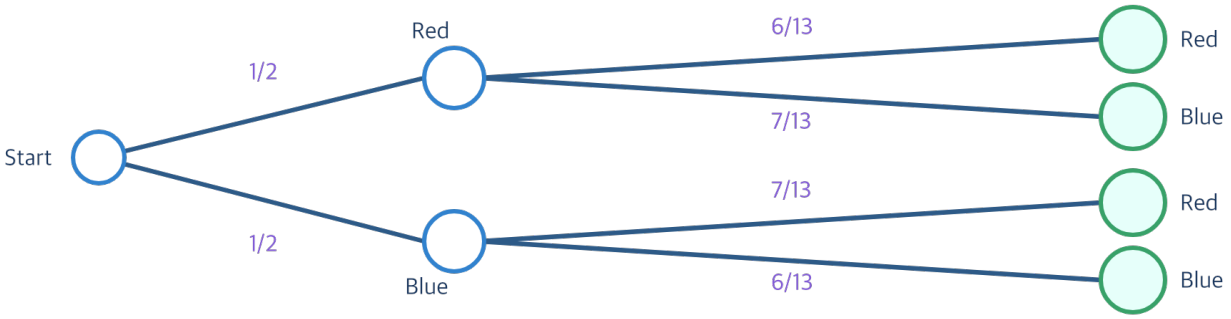
Lesson starter

Tree Diagrams and Conditional Probability

Conditional Trees

Name _____

SB1



A bag contains 7 red counters and 7 blue counters. Two counters are drawn without replacement. Find P(one red and one blue in any order).

STA-3EBB-L2 · SB

Show your working

KAIZEN MATHS

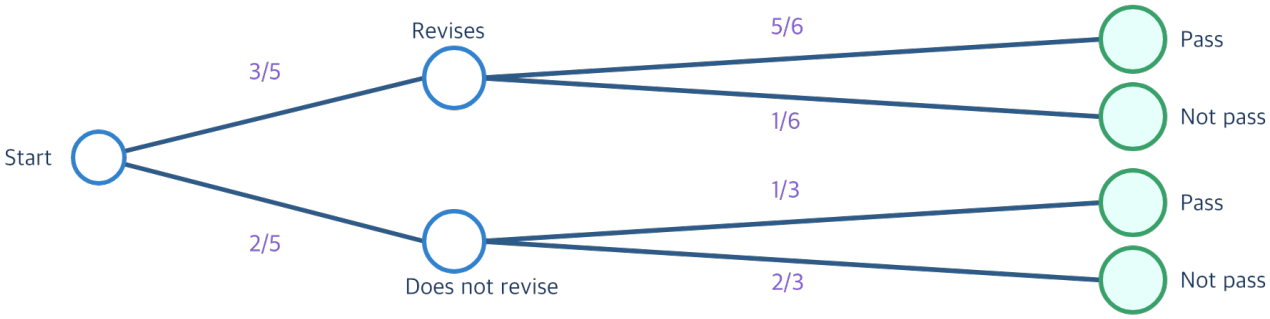
Lesson starter

Tree Diagrams and Conditional Probability

Conditional Trees

Name _____

SC1



The tree shows a situation where P(Revises) = $\frac{3}{5}$, P(Pass | Revises) = $\frac{5}{6}$, and P(Pass | Does not revise) = $\frac{1}{3}$. Find P(Pass).

STA-3EBB-L2 · SC

Show your working

KAIZEN MATHS

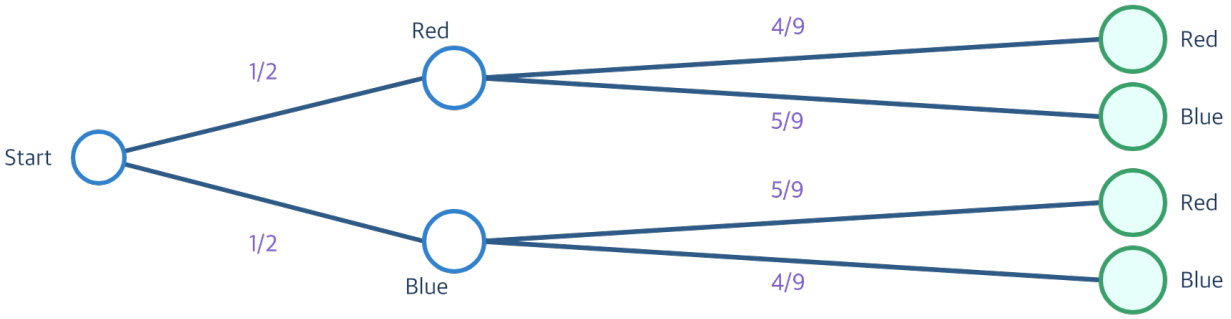
Lesson starter

Tree Diagrams and Conditional Probability

Conditional Trees

Name _____

SD1



A bag contains 5 red counters and 5 blue counters. Two counters are drawn without replacement. Find P(two blue counters).

STA-3EBB-L2 · SD

Show your working

KAIZEN MATHS

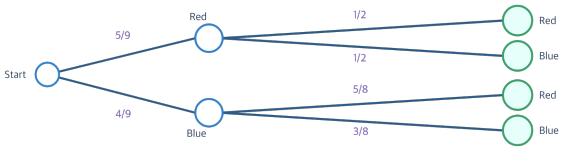
Exit ticket

Tree Diagrams and Conditional Probability

Conditional Trees

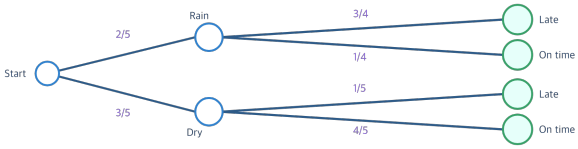
Name _____

EA1



A bag contains 5 red counters and 4 blue counters. Two counters are drawn without replacement. Find P(one red and one blue in any order).

EA2



The tree shows a situation where P(Rain) = $\frac{2}{5}$, P(Late | Rain) = $\frac{3}{4}$, and P(Late | Dry) = $\frac{1}{5}$. Find P(Late).

Show your working

KAIZEN MATHS

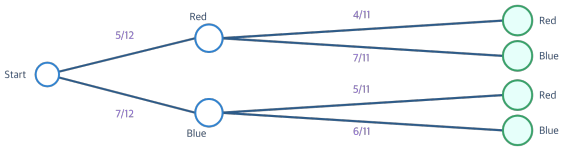
Exit ticket

Tree Diagrams and Conditional Probability

Conditional Trees

Name _____

EB1



A bag contains 5 red counters and 7 blue counters. Two counters are drawn without replacement. Find P(two blue counters).

Show your working

KAIZEN MATHS

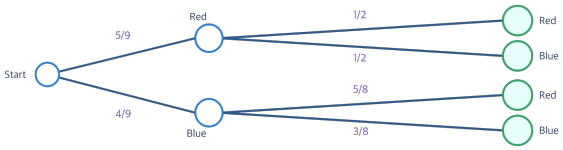
Exit ticket

Tree Diagrams and Conditional Probability

Conditional Trees

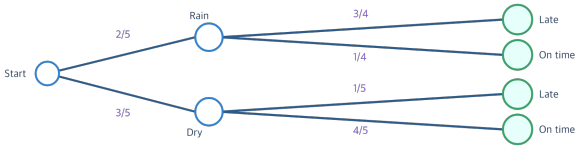
Name _____

EA1



A bag contains 5 red counters and 4 blue counters. Two counters are drawn without replacement. Find P(one red and one blue in any order).

EA2



The tree shows a situation where P(Rain) = $\frac{2}{5}$, P(Late | Rain) = $\frac{3}{4}$, and P(Late | Dry) = $\frac{1}{5}$. Find P(Late).

STA-3EBB-L2 · EA

Show your working

KAIZEN MATHS

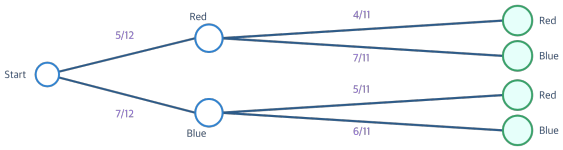
Exit ticket

Tree Diagrams and Conditional Probability

Conditional Trees

Name _____

EB1



A bag contains 5 red counters and 7 blue counters. Two counters are drawn without replacement. Find P(two blue counters).

STA-3EBB-L2 · EB

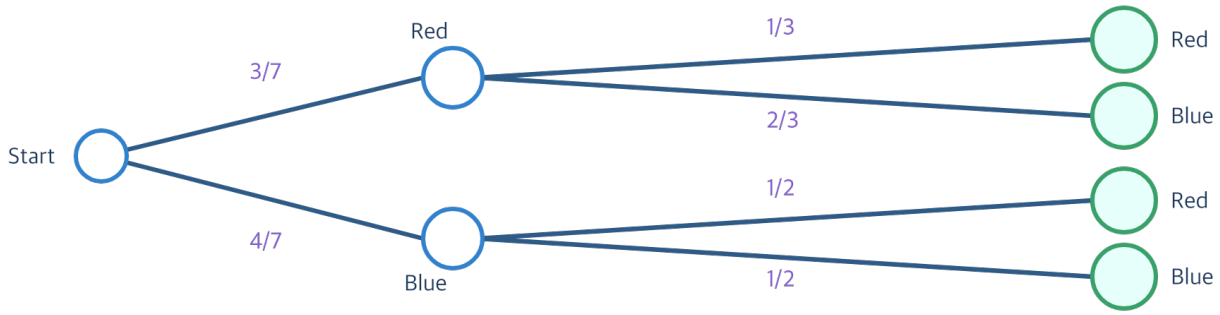
Show your working

Teacher answers & working

Tree Diagrams and Conditional Probability · Conditional Trees

STA-3EBB-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



A bag contains 3 red counters and 4 blue counters. Two counters are drawn without replacement. Find P(two blue counters).

Answer:

$\frac{2}{7}$

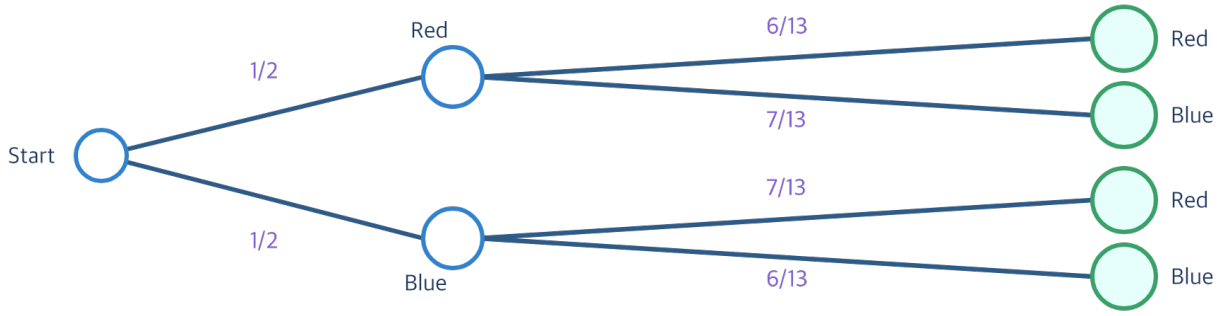
Working:

Without replacement means the second draw depends on the first draw.

P(Blue then Blue) = $\frac{4}{7} \times \frac{1}{2} = \frac{2}{7}$

$\frac{2}{7} = 28.6\%$

SB1



A bag contains 7 red counters and 7 blue counters. Two counters are drawn without replacement. Find P(one red and one blue in any order).

Answer:

$\frac{7}{13}$

Working:

One of each can happen as Red then Blue or Blue then Red.

Red then Blue: $\frac{1}{2} \times \frac{7}{13} = \frac{7}{26}$

Blue then Red: $\frac{1}{2} \times \frac{7}{13} = \frac{7}{26}$

Add the two paths: $\frac{7}{26} + \frac{7}{26} = \frac{7}{13}$

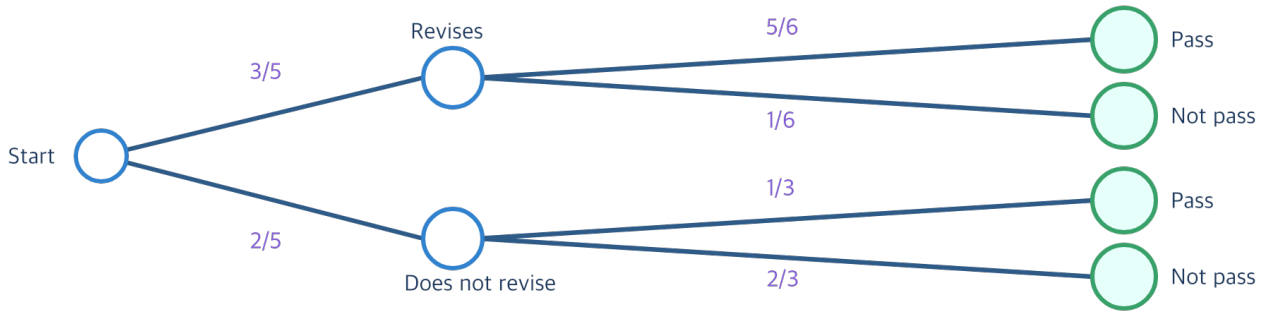
$\frac{7}{13} = 53.8\%$

Teacher answers & working

Tree Diagrams and Conditional Probability · Conditional Trees

STA-3EBB-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



The tree shows a situation where P(Revises) = $\frac{3}{5}$, P(Pass | Revises) = $\frac{5}{6}$, and P(Pass | Does not revise) = $\frac{1}{3}$. Find P(Pass).

Answer:

$\frac{19}{30}$

Working:

To get Pass, add every path that ends in Pass.

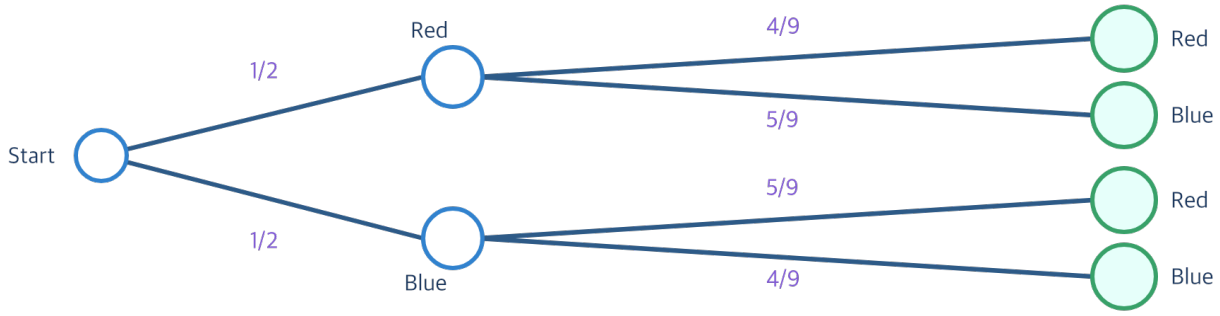
Revises and Pass: $\frac{3}{5} \times \frac{5}{6} = \frac{1}{2}$

Does not revise and Pass: $\frac{2}{5} \times \frac{1}{3} = \frac{2}{15}$

P(Pass) = $\frac{1}{2} + \frac{2}{15} = \frac{19}{30}$

$\frac{19}{30} = 63.3\%$

SD1



A bag contains 5 red counters and 5 blue counters. Two counters are drawn without replacement. Find P(two blue counters).

Answer:

$\frac{2}{9}$

Working:

Without replacement means the second draw depends on the first draw.

P(Blue then Blue) = $\frac{1}{2} \times \frac{4}{9} = \frac{2}{9}$

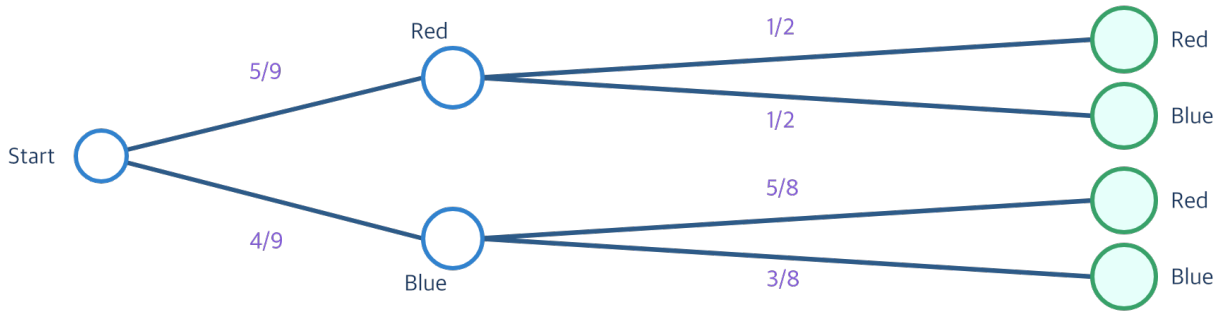
$\frac{2}{9} = 22.2\%$

Teacher answers & working

Tree Diagrams and Conditional Probability · Conditional Trees

STA-3EBB-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



A bag contains 5 red counters and 4 blue counters. Two counters are drawn without replacement. Find P(one red and one blue in any order).

Answer:

$\frac{5}{9}$

Working:

One of each can happen as Red then Blue or Blue then Red.

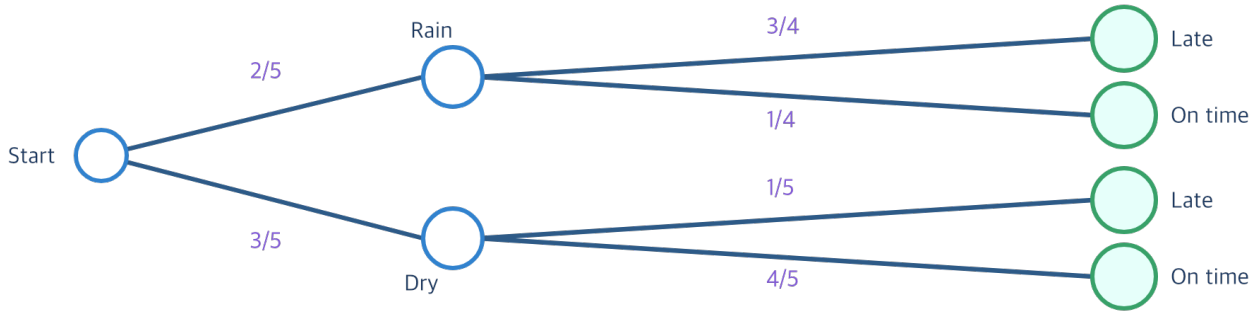
Red then Blue: $\frac{5}{9} \times \frac{1}{2} = \frac{5}{18}$

Blue then Red: $\frac{4}{9} \times \frac{5}{8} = \frac{5}{18}$

Add the two paths: $\frac{5}{18} + \frac{5}{18} = \frac{5}{9}$

$\frac{5}{9} = 55.6\%$

EA2



The tree shows a situation where P(Rain) = $\frac{2}{5}$, P(Late | Rain) = $\frac{3}{4}$, and P(Late | Dry) = $\frac{1}{5}$. Find P(Late).

Answer:

$\frac{21}{50}$

Working:

To get Late, add every path that ends in Late.

Rain and Late: $\frac{2}{5} \times \frac{3}{4} = \frac{3}{10}$

Dry and Late: $\frac{3}{5} \times \frac{1}{5} = \frac{3}{25}$

P(Late) = $\frac{3}{10} + \frac{3}{25} = \frac{21}{50}$

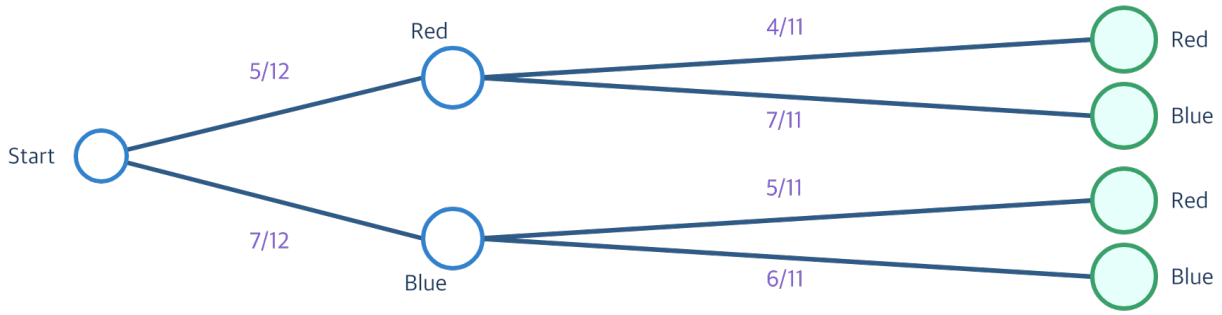
$\frac{21}{50} = 42.0\%$

Teacher answers & working

Tree Diagrams and Conditional Probability · Conditional Trees

STA-3EBB-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



A bag contains 5 red counters and 7 blue counters. Two counters are drawn without replacement. Find P(two blue counters).

Answer:

$\frac{7}{22}$

Working:

Without replacement means the second draw depends on the first draw.

P(Blue then Blue) = $\frac{7}{12} \times \frac{6}{11} = \frac{7}{22}$

$\frac{7}{22} = 31.8\%$